

HDOC DAY : 31

Aim

To write a menu-driven C program to perform operations on a 2D array (matrix):

1. Calculate the sum of all elements in the matrix.
2. Find the sum of each row and store the sums in a 1D array.
3. Find and display the transpose of the matrix.
4. Exit the menu loop.

Algorithm

1. Start the program.
2. Read the number of rows (r) and columns (c) of the matrix from the user.
3. Read the elements of the 2D array (matrix) using nested for loops.
4. Display a menu with options:
 - a. **Option 1:** Sum of all elements
 - b. **Option 2:** Sum of each row
 - c. **Option 3:** Transpose of the matrix
 - d. **Option 4:** Exit
5. Ask the user to enter their choice.
6. **If choice is 1:**
 - a. Initialize a variable sum = 0.
 - b. Loop through each element of the matrix and add it to sum.
 - c. Print the total sum.
7. **If choice is 2:**
 - a. Create a 1D array rowSum of size r.
 - b. Loop through each row i. Initialize rowSum[i] = 0.
 - c. Loop through each column j and add the matrix elements to rowSum[i].

d. Print the sum of each row and the final rowSum array.

8. If choice is 3:

- a. Loop through the matrix columns first, then rows.
- b. Print the elements in inverted index order (i.e., matrix[j][i]) to display the transpose.

9. If choice is 4:

- a. Stop the program.
10. Repeat steps 4–9 using a do-while loop until the user selects choice 4.

C CODE :

```
#include <stdio.h>
```

```
int main() {
```

```
    int r, c, i, j, choice;
```

```
    int a[10][10];
```

```
    int rowSum[10];
```

```
    int total;
```

```
    printf("Enter the number of rows and columns: ");
```

```
    scanf("%d %d", &r, &c);
```

```
    printf("Enter the matrix elements:\n");
```

```
    for (i = 0; i < r; i++) {
```

```
        for (j = 0; j < c; j++) {
```

```
            scanf("%d", &a[i][j]);
```

```
}  
}
```

```
while (1) {  
    printf("\nEnter your choice: ");  
    scanf("%d", &choice);  
  
    if (choice == 1) {  
        total = 0;  
        for (i = 0; i < r; i++) {  
            for (j = 0; j < c; j++) {  
                total = total + a[i][j];  
            }  
        }  
        printf("Sum of all elements = %d\n", total);  
    }  
    else if (choice == 2) {  
        printf("Row sums:\n");  
        for (i = 0; i < r; i++) {  
            rowSum[i] = 0;  
            for (j = 0; j < c; j++) {  
                rowSum[i] = rowSum[i] + a[i][j];  
            }  
        }  
    }  
}
```

```

        printf("Sum of row %d = %d\n", i + 1, rowSum[i]);
    }

    printf("The row-sum array contains: [");
    for (i = 0; i < r; i++) {
        printf("%d", rowSum[i]);
        if (i < r - 1) {
            printf(", ");
        }
    }
    printf("]\n");
}

else if (choice == 3) {
    printf("Transpose of the matrix:\n");
    for (i = 0; i < c; i++) {
        for (j = 0; j < r; j++) {
            printf("%d ", a[j][i]);
        }
        printf("\n");
    }
}

else if (choice == 4) {
    break;
}

```

```
    }  
}  
  
return 0;  
}
```

CODE SCREENSHOT :

```
#include <stdio.h>

int main() {
    int r, c, i, j, choice;
    int a[10][10];
    int rowSum[10];
    int total;

    printf("Enter the number of rows and columns: ");
    scanf("%d %d", &r, &c);

    printf("Enter the matrix elements:\n");
    for (i = 0; i < r; i++) {
        for (j = 0; j < c; j++) {
            scanf("%d", &a[i][j]);
        }
    }

    while (1) {
        printf("\nEnter your choice: ");
        scanf("%d", &choice);

        if (choice == 1) {
            total = 0;
            for (i = 0; i < r; i++) {
                for (j = 0; j < c; j++) {
                    total = total + a[i][j];
                }
            }
            printf("Sum of all elements = %d\n", total);
        }
        else if (choice == 2) {
            printf("Row sums:\n");
            for (i = 0; i < r; i++) {
                rowSum[i] = 0;
                for (j = 0; j < c; j++) {
                    rowSum[i] = rowSum[i] + a[i][j];
                }
                printf("Sum of row %d = %d\n", i + 1, rowSum[i]);
            }

            printf("The row-sum array contains: [");
            for (i = 0; i < r; i++) {
                printf("%d", rowSum[i]);
                if (i < r - 1) {
                    printf(", ");
                }
            }
            printf("]\n");
        }
        else if (choice == 3) {
            printf("Transpose of the matrix:\n");
            for (i = 0; i < c; i++) {
                for (j = 0; j < r; j++) {
                    printf("%d ", a[j][i]);
                }
            }
        }
    }
}
```

```
        printf("%d ", a[j][i]);  
    }  
    printf("\n");  
}  
}  
else if (choice == 4) {  
    break;  
}  
}  
  
return 0;  
}
```

OUTPUT SCREENSHOT :

```
[manalithakar@Manalis-MacBook-Air ~ % nano 31.c
[manalithakar@Manalis-MacBook-Air ~ % gcc 31.c -o 31
[manalithakar@Manalis-MacBook-Air ~ % ./31
Enter the number of rows and columns: 3 2
Enter the matrix elements:
10 20
30 40
50 60

Enter your choice: 1
Sum of all elements = 210

Enter your choice: 2
Row sums:
Sum of row 1 = 30
Sum of row 2 = 70
Sum of row 3 = 110
The row-sum array contains: [30, 70, 110]

Enter your choice: 3
Transpose of the matrix:
10 30 50
20 40 60

Enter your choice: 4
manalithakar@Manalis-MacBook-Air ~ %
```